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STRETCHING UNIT AND FILM PRODUCTION SYSTEM

Abstract

A stretching unit for stretching a film, in particular a transverse direction orienter and/or simultaneous stretching unit, including an oven and a heat recovery system is disclosed. The heat recovery system includes a heat exchanger with an exhaust air duct for exhaust air from the oven, a supply air duct for supply air to the oven, a plurality of heat conductors, a plurality of heat collectors and a condensate collector. The exhaust air duct is open towards the condensate collector, in which the heat collectors are located in the exhaust air duct with an interstice between adjacent heat collectors, and the heat conductors are connected thermally to the heat collectors, and in which the interstices are open towards the condensate collector.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to German Patent Application Number DE 10 2024 106 937.6, filed Mar. 11, 2024, the entire contents of which is hereby incorporated by reference.

BACKGROUND

[0002] The disclosure relates to a stretching unit as well as a film production system.

[0003] Film production systems are known and usually comprise at least one stretching unit.

[0004] The production of plastic film in such systems is very energy-intensive as high temperatures are required in the ovens of the stretching units.

[0005] At that same time, fresh air must be fed to the oven continuously and in return exhaust air must be extracted in order to reduce contamination in the oven. Contamination occurs, among other things, as different kinds of substances, primarily hydrocarbons, escape from the film when passing through the oven. The air exchange here is related to high heating energy demands as the fresh air drawn in must be heated to the oven temperature.

[0006] It is thus known to provide film production systems with a heat exchanger for heat recovery. However, it is problematic in such systems that the substances contained in the oven air condense on the heat exchanger when cooling and precipitate on the exhaust-air side of the surface of the heat exchanger. Consequently, the efficiency of the heat exchanger deteriorates so that the heat exchanger must be cleaned laboriously.

SUMMARY

[0007] It is provided a stretching unit and a film production system that is particularly energy-efficient and simultaneously low-maintenance.

[0008] A stretching unit for stretching a film, in particular transverse direction orienter and/or simultaneous stretching unit, is provided comprising an oven and a heat recovery system. The heat recovery system comprises a condensate collector and a heat exchanger with an exhaust air duct for exhaust air from the oven, a supply air duct for supply air to the oven, a plurality of heat conductors and a plurality of heat collectors. The supply air duct and the exhaust air duct are separated fluidly from each other by a partition, and the heat conductors extend from the exhaust air duct through the partition into the supply air duct, wherein the exhaust air duct is open towards the condensate collector. The heat collectors are located in the exhaust air duct with an interstice between adjacent heat collectors, and the heat conductors are connected thermally to the heat collectors, wherein the interstices are open towards the condensate collector.

[0009] A higher degree of efficiency of the heat exchanger is achieved by means of the heat collectors. At the same time, the precipitated condensate is transported away reliably through the condensate collector and as the interstices are open towards the condensate collector. Consequently, the high degree of efficiency also remains permanently high without cleaning and thus maintenance.

[0010] For example, each heat collector is connected thermally to at least one, multiple or all of the heat conductors.

[0011] The supply air is, in particular, fresh air from the environment of the stretching unit.

[0012] In an embodiment, multiple heat emitters that are designed like the heat collectors are provided in the supply air duct.

[0013] In an embodiment, the exhaust air duct defines a flow path for the air flow of the exhaust air, wherein the condensate collector is located outside of the flow path. In this way, it avoids the condensate being carried from the heat exchanger by the air flow.

[0014] For a further improved removal of the condensate from the interstices, the condensate collector can be arranged at least in part below the exhaust air duct, in particular below the flow

path.

[0015] Within the scope of this disclosure, the term “below” is understood to mean in relation to the provided assembly position of the heat recovery system. In particular, “below” is to mean directly vertically below and not additionally horizontally offset.

[0016] In an embodiment, the exhaust air duct comprises a condensate outlet opening that is part of the condensate collector, in particular wherein the exhaust air duct comprises a base and said at least one condensate outlet opening is provided in the base. In this way, the collected condensate can be removed reliably from the exhaust air duct.

[0017] For example, the base is designed level or as a channel.

[0018] In an embodiment, the condensate collector comprises a tray, in particular wherein the tray is removeable from the condensate collector for emptying. As a result, the condensate can be collected reliably and, in particular, disposed of simply.

[0019] To reduce maintenance further, the heat recovery system can comprise a condensate drain and the condensate collector is connected fluidly to the condensate drain, in particular by means of a siphon.

[0020] To improve the efficiency of the heat exchanger further, the heat collectors can be designed plate-shaped and the heat conductors can extend through the heat collectors.

[0021] For example, each heat conductor extends through at least one, multiple or all heat collectors, in particular perpendicularly.

[0022] In an embodiment, the heat collectors run parallel to each other and/or parallel to the partition in order to reduce flow resistance.

[0023] To drain away any condensate at the collectors reliably, the heat conductors can run horizontally or be arranged at an angle of between 0° and 10° to the horizontal direction, in particular wherein the section of the heat conductor in the exhaust air duct is situated lower than that section in the supply air duct.

[0024] For example, the angle is 4° .

[0025] In an embodiment, the interstices comprise a direction of air flow and a direction of condensate flow, wherein the direction of air flow and the direction of condensate flow run parallel to each other, in particular wherein the direction of air flow and the direction of condensate flow run vertically or are orientated at an angle of between 0° and 10° to the vertical. In this way, the removal of the condensate from the interstices is supported by the air flow.

[0026] The direction of condensate flow is, for example, that direction in which the condensate formed at the heat collectors can flow unimpeded—except for the heat conductors. The direction of condensate flow runs in particular downwards.

[0027] The heat conductors can extend in the region of the heat collectors perpendicular to the direction of air flow and/or the direction of condensate flow.

[0028] In an embodiment, the interstices comprise a direction of air flow and a direction of condensate flow, wherein the direction of air flow and the direction of condensate flow run perpendicular to each other, in particular wherein the direction of air flow runs horizontally and/or the direction of condensate flow runs vertically or is orientated at an angle of between 0° and 10° to the vertical. In this way, the amount of condensate being transported into the environment through the air flow is reduced further.

[0029] To increase efficiency further, the heat conductors can be heat-transfer devices, in particular heat pipes.

[0030] In an embodiment, the oven comprises a supply air inlet which is connected fluidly to the supply air duct of the heat exchanger, and/or the oven comprises an exhaust air outlet that is connected fluidly to the exhaust air duct of the heat exchanger so that a direct and efficient connection of the heat recovery system to the oven is achieved.

[0031] The fluid connection occurs, for example, by means of piping.

[0032] To temperature control the supply air, the heat recovery system can comprise a heating

register. For example, the heating register can be arranged fluidly between the supply air duct of the heat exchanger and the supply air inlet of the oven.

[0033] The supply air fan is located, for example, between the supply air duct of the heat exchanger and/or of the heating register and the supply air inlet of the oven. The exhaust air fan is located, for example, downstream of the exhaust air duct of the heat exchanger.

[0034] There is also provided a production system comprising at least one stretching unit as described previously as well as one further stretching unit, an extrusion unit, a cast rolling system, a draw roller unit and/or a winder unit.

[0035] The features and advantages described for the stretching unit apply equally to the film production system and vice versa.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0036] Additional features and advantages of the disclosure are found in the following description as well as the attached drawings to which reference is made. In the drawings:

[0037] FIG. 1 shows a film production system according to an embodiment of the disclosure comprising at least a stretching unit according to an embodiment of the disclosure in a schematic, perspective view,

[0038] FIG. 2 shows an enlarged representation of a heat recovery system of the stretching unit according to FIG. 1,

[0039] FIGS. 3, 4 show schematic sectional views through the heat exchanger of the heat recovery system according to FIG. 2,

[0040] FIG. 5 shows a perspective view of a heat exchanger of a stretching unit according to a second embodiment of the disclosure, and

[0041] FIG. 6 shows a sectional view through the heat exchanger of a heat recovery system of a stretching unit according to a third embodiment of the disclosure.

DETAILED DESCRIPTION

[0042] Lists having a plurality of alternatives connected by “and/or”, for example “A, B and/or C” are to be understood to disclose an arbitrary combination of the alternatives, i.e. the lists are to be read as “A and/or B and/or C” or as “at least one of A, B or C”. The same holds true for listings with more than three items.

[0043] In FIG. 1, a film production system **10** is shown extremely schematically, which comprises several different units and devices.

[0044] In the shown example, the film production system **10** comprises an extrusion unit **12**, a cast rolling system **14**, at least one stretching unit **15**—such as a machine direction orienter **16** (MDO) or a transverse direction orienter **18** (TDO)—, a draw roller unit **20** as well as a winder unit **22**.

[0045] The film produced is, for example, a biaxially stretched film, such as polypropylene film (BOPP), PP capacitor film (BOPP-C), polyethylene terephthalate film (BOPET), polyamide film (BOPA), polyethylene film (BOPE) or polylactic acid film (BOPLA).

[0046] To produce plastic films, a film is created on the chill roll of a cast rolling system **14** by means of an extrusion unit **12**. To this end, the extrusion unit **12** generates a melt from the starting products, such as granular material, said melt being applied to the chill roll, thereby creating the film.

[0047] This film is conveyed from the cast rolling system **14** to the machine direction orienter **16**. In the machine direction orienter **16**, the film is stretched in the machine direction in order to obtain a stretched film.

[0048] In the machine direction orienter **16**, the film runs over a plurality of rollers that are heated in order heat the film to the desired temperature so that it can be stretched.

[0049] Between at least two of the rollers present in the machine direction orienter **16**, the stretching takes place in the machine direction, i.e. in the drawing direction, so that the film becomes a stretched film.

[0050] The film obtained is conveyed by the machine direction orienter **16** to the transverse direction orienter **18** and stretched in the transverse direction orienter **18** in the transverse direction.

[0051] Along the drawing direction of the film production system **10**, the transverse direction orienter **18** has an oven **26** comprising different zones for treating the film.

[0052] The film is heated in the first zone, also termed the preheating zone. In the subsequent second zone (“stretching zone”), the film is stretched in the transverse direction so that its width is greater and its thickness is less at the end of the second zone than it was at the start.

[0053] After completing the stretching, the film then passes through the third and further zones (termed “heat treatment zone”, “further heating zone” and/or “annealing zone”), in which a relaxation of the film, for example, can take place at high temperatures.

[0054] Subsequently, the film passes through a further zone (“cooling zone”), whereby the film is cooled in the last zone.

[0055] A further zone is termed the neutral zone and serves to separate the zones. The neutral zone is, for example, an empty space without any ventilation.

[0056] The zones of the transverse direction orienter **18** can also be divided differently and/or designed in their lengths differently. For example, fewer or shorter neutral zones can be provided or the neutral zones can be arranged at other points, also additionally. Changes in the remaining zones are also conceivable.

[0057] After the transverse direction orienter **18**, the now biaxially stretched film runs through the draw roller unit **20** and is wound by means of the winder unit **22**.

[0058] It is also conceivable that the film production system **10** is designed in another way, for example comprising a simultaneous stretching unit **19** with an oven **26** as a stretching unit **15** alternatively or in addition to the machine direction orienter **16** and/or to the transverse direction orienter **18**.

[0059] Within the scope of this disclosure, a stretching unit **15** is now described in general that can both be a transverse direction orienter or simultaneous stretching unit or a combination of these.

[0060] The stretching unit **15**, more specifically its oven **26**, continually requires supply air, particularly fresh air, during operation in order to be capable of extracting contaminated exhaust air in return.

[0061] To be capable of using the exhaust heat from the exhaust air, the stretching unit **15** comprises a heat recovery system **28**.

[0062] In FIG. **2**, one of the heat recovery systems **28** is enlarged, of which a stretching unit **15** can also comprise several ones.

[0063] It can be seen that the oven **26** comprises a supply air inlet **30** and a exhaust air outlet **32**, which are both connected fluidly to the heat recovery system **28**.

[0064] The heat recovery system **28** comprises a heat exchanger **34**, a supply air fan **36**, a exhaust air fan **38**, a heating register **40**, piping **42** as well as a condensate collector **52**.

[0065] The heat exchanger **34** has a supply air duct **44** as well as an exhaust air duct **46** that are fluidly separate from each other and each have a fluid connection to the supply air inlet **30** or the exhaust air outlet **32**, respectively, of the oven **26** by means of the piping **44**.

[0066] The supply air duct **44** of the heat exchanger **34** is open upstream in relation to the supply air flow towards the environment of the oven **26** and the heat recovery system **28**. Optionally, an air filter is provided on this side.

[0067] Downstream, the supply air duct **44** of the heat exchanger **34** is connected fluidly to the heating register **40** and the supply air fan **36**, which are arranged between the supply air duct **44** and the supply air inlet **30**. For example, the supply air fan **36** is arranged downstream of the heating register **40** so that the heating register **40** is located between the heat exchanger **34** and the

supply air fan **36**.

[0068] The exhaust air fan **38** can be located, in relation to the exhaust air flow, downstream of the exhaust air duct **46** of the heat exchanger **34** so that the heat exchanger **34** is located fluidly between the exhaust air fan **38** and the exhaust air outlet **32** of the oven **26**.

[0069] It is also conceivable that the exhaust air fan **38** is located between the exhaust air outlet **32** and the heat exchanger **34**.

[0070] In FIGS. **3** and **4**, the heat exchanger **34** of the heat recovery system **28** is shown simplified in the longitudinal view and in the cross section (each orthogonally to the direction of flow).

[0071] In addition to the supply air duct **44** and the exhaust air duct **46**, the heat exchanger **34** comprises a plurality of heat conductors **48**, a plurality of heat collectors **50** as well as a partition **54**.

[0072] For the sake of clarity, only a few heat conductors **48** and a few heat collectors **50** are shown in the figures in order to explain the basic principle. It is conceivable that considerably more heat conductors **48** and/or heat collectors **50** are provided.

[0073] In FIGS. **3** and **4**, the heat exchanger **34** is shown in the provided assembly position so that parts shown below in relation to gravity are also arranged below in the figures.

[0074] In the heat exchanger **34**, the supply air duct **44** and the exhaust air duct **46** are located directly adjacent to each other. The supply air duct **44** is separated fluidly from the exhaust air duct **46** by means of a partition **54**.

[0075] The exhaust air duct **46** defines a flow path for the air flow S.sub.A of the exhaust air. The direction of air flow S.sub.A of the exhaust air in the exhaust air duct **46** and the direction of air flow S.sub.Z of the supply air in the supply air duct **44** run antiparallel to each other. It is conceivable that the direction of air flow S.sub.A of the exhaust air in the exhaust air duct **46** and the direction of air flow S.sub.Z of the supply air in the supply air duct **44** run parallel to each other.

[0076] In the first embodiment, the direction of air flow S.sub.A in the exhaust air duct **46** and the direction of air flow S.sub.Z of the supply air duct **44** run horizontally.

[0077] The condensate collector **52** is located below the exhaust air duct **46**, and the exhaust air duct **46** is open towards the condensate collector **52**. The condensate collector **52** is located outside, in particular below the flow path.

[0078] The condensate collector **52** can be realised at least in part, in particular completely, through condensate outflow openings **56** that are provided in the base **58** of the exhaust air duct **46**. The base **58** can be designed here as a channel.

[0079] Below means in this regard in particular that at least parts of the condensate collector **52** or the entire condensate collector **52** is located directly vertically below at least parts of the exhaust air duct **46** or the entire exhaust air duct **46**, i.e. following the force of gravity.

[0080] The condensate collector **52** comprises in addition a tray **60** that is located, for example, below the condensate outflow openings **56**. To drain the tray **60**, the heat recovery system **28** can comprise a condensate drain **62**, as shown in FIG. **3** with dashed lines. The condensate drain **62** is connected fluidly to the condensate collector **52**, in FIG. **3** to the tray **60**, for example by means of a siphon **64**.

[0081] The heat conductors **48** are designed as heat-transfer devices, for example, in particular as heat pipes.

[0082] The heat conductors **48** are each located in part in the supply air duct **44** and in part in the exhaust air duct **46**. Thus, they extend from the exhaust air duct **46** through the partition **54** into the supply air duct **44**. In relation to the directions of air flow S.sub.A and S.sub.Z, the heat conductors **48** run perpendicularly.

[0083] The heat conductors **48** can run in the horizontal direction H or, as shown in FIG. **3**, at an angle α to the horizontal direction H of 0° to 10° . For example, the angle is 4° . Here, the heat conductors **48** ascend towards the supply air duct **44** so that the section of each heat conductor **48**

in the exhaust air duct **46** is situated lower than the section of the same heat conductor **48** in the supply air duct **44**.

[0084] The heat collectors **50** are located in the exhaust air duct **46** and designed plate-shaped in the shown embodiment. They are arranged parallel to each other and run, for example, parallel to the partition **54**.

[0085] The heat collectors **50** extend in the direction of air flow S.sub.A as well as in the vertical direction V or at an angle β of between 0° and 10° to the vertical direction V.

[0086] The heat collectors **50** extend perpendicular to the heat conductors **48**, wherein they are connected thermally to one, multiple or all heat conductors **48**. One, multiple or all heat conductors extend, for example, through one of the heat collectors **50** and are connected to it thermally.

[0087] Similarly, each of the heat conductors **48** are connected thermally to one, multiple or all heat collectors **50**.

[0088] In the same way as the heat collectors **50**, a plurality of heat emitters **66** are provided in the supply air duct **44**, wherein only one of the heat emitters **66** is shown in FIG. 3 for the sake of clarity.

[0089] The heat collectors **50** are located parallel to each other so that an interstice **68** is formed in each case between two adjacent heat collectors **50**. The interstices **68** thus run in the same directions as the heat collectors **50**.

[0090] The interstices **68** are open in the direction of air flow S.sub.A as a result of the arrangement of the heat collectors. Thus, the direction of air flow S.sub.A, in which the exhaust air flows into the interstices **68**, can also be allocated to the interstices **68**.

[0091] Similarly, the interstices **68** are open towards the condensate collector **52**, i.e. downwards.

[0092] The interstices **68** comprise a direction of condensate flow F that describes that direction in which the condensate precipitating from the exhaust air on the heat collectors **50** flows off. The direction of condensate flow F runs, for example, downwards and has a component directed primarily downwards, i.e. in the direction of gravity.

[0093] The interstices **68** also extend like the heat collectors **50** in the vertical direction V or at the angle β to the vertical direction V. In the example of FIG. 3, the heat conductors **48** extend perpendicular to the direction of condensate flow F.

[0094] In the example according to FIG. 3, the direction of condensate flow F runs perpendicular to the direction of air flow S.sub.A.

[0095] During the operation of the film production system **10** and the stretching unit **15** and thus also the heat recovery system **28**, warm air is conveyed from the oven **26** through the exhaust air duct **46** by means of the exhaust air fan **38**.

[0096] This exhaust air contains, for example, hydrocarbons that condense rapidly when cooling.

[0097] The warm exhaust air thus flows through the exhaust air duct **46** of the heat exchanger **34** in the direction of air flow S.sub.A and there, comes into contact with the heat collectors **50** and the heat conductors **48**. Subsequently, the exhaust air is transported through the exhaust air fan **38** into the environment of the stretching unit **15**.

[0098] In the exhaust air duct **46**, the exhaust air transfers heat to the heat collectors **50**, which are colder in comparison to the exhaust air, and the heat conductors **48**.

[0099] This emitted heat is transported through the heat collectors **50** and then through the heat conductors **48** into the supply air duct **44**.

[0100] Supply air, in particular fresh air, conveyed by means of the supply air fan **36** flows through the supply air duct **44**.

[0101] In the supply air duct **44**, the supply air is heated due to contact to the heat conductors **48** and the heat emitters **66** that are warmer than compared to the supply air. Subsequently, the supply air passes the heating register **40** and is heated to the temperature required in the oven until it enters the oven **26** through the supply air inlet **30**.

[0102] In the exhaust air duct **46**, as described, the exhaust air is cooled at the heat collectors **50**.

The hydrocarbons found in the air condense as a result of this cooling and form a condensate on the heat collectors **50** and the heat conductors **48**.

[0103] Owing to the arrangement of the heat collectors **50** and the interstices **68**, the condensate can flow downwards, i.e. in the direction of condensate flow **F**, without any impediment—except for the traversing heat conductors **48**.

[0104] As soon as the condensate reaches the lower end of the heat collectors **50**, it can drain or flow away from the heat collectors **50** and is collected by the condensate collector **52**.

Subsequently, the condensate can be disposed of through the condensate drain **62**.

[0105] In this way, it ensures that no condensate remains in the exhaust air duct **46** and specifically in the interstices **68** and that these can be clogged. In this way, a very good degree of efficiency of the heat exchanger **34** is ensured without complex maintenance activities being necessary.

[0106] The heat recovery system **28** is thus extremely efficient and at the same low-maintenance.

[0107] In FIGS. **5** and **6**, further embodiments of the heat recovery system **28** and thus the stretching unit **15** and the film production system **10** are shown. These correspond substantially to the first embodiment so that only the differences are discussed hereinafter and the same and functionally equivalent parts are provided with the same reference signs.

[0108] FIG. **5** shows a perspective view of a part of a second embodiment of a heat exchanger **34**.

[0109] In this embodiment, the tray **60** of the condensate collector **52** can be removed, i.e. removed from below the exhaust air duct **46**. In this way, the tray **60** can be emptied easily without needing to provide a dedicated condensate drain **62**.

[0110] In FIG. **5**, the condensate outflow openings **56** are also clearly evident in the base **58** of the exhaust air duct **46**.

[0111] For example, the condensate outflow openings **56** are provided on the downstream edge of the base **58** so that the air flow **S.sub.A** conveys the condensate drained on the base to the condensate outflow openings **56**.

[0112] In FIG. **6**, a third embodiment of the heat recovery system **28** is shown.

[0113] In this embodiment, the direction of air flow **S.sub.A**, **S.sub.Z** of the exhaust air or the supply air do not run horizontally, but vertically towards the heat collectors **50** and the heat conductors **48**.

[0114] Below the heat collectors **50** and the heat conductors **48**, the exhaust air duct **46** runs horizontally and/or offset from the heat exchanger **34** so that the flow path also accordingly changes direction (indicated by arrows in FIG. **6**).

[0115] The condensate collector **52** is located below the site at which the flow path carries out the change in direction.

[0116] The condensate collector can be designed as described for the previous embodiments.

[0117] The arrangement of the heat collectors **50** and the heat conductors **48** in the supply air duct **44** correspond to the arrangement in the first embodiment so that the interstices **68** are still open towards the condensate collector **52**.

[0118] In this embodiment, the direction of air flow **S.sub.A** and the direction of condensate flow **F** now run parallel to each other within the region of the heat collectors **50**.

[0119] The direction of air flow **S.sub.A** and the direction of condensate flow **F** run in the vertical direction **V** or, as shown, at an angle β of between 0° and 10° to the vertical direction **V**.

[0120] The removal of condensate from the interstices **68** is improved in this embodiment as **10** the direction of air flow **S.sub.A** and the direction of condensate flow **F** run in a parallel manner.

[0121] The different features of various embodiments are combinable with each other, in particular the use of a removeable tray **60**, a condensate drain **62** and/or the position of the condensate outflow openings **56** in the base **58**.

Claims

- 1.** A stretching unit for stretching a film, comprising: an oven and a heat recovery system, wherein the heat recovery system comprises a condensate collector and a heat exchanger with an exhaust air duct for exhaust air from the oven, a supply air duct for supply air to the oven, a plurality of heat conductors and a plurality of heat collectors, wherein the supply air duct and the exhaust air duct are separated fluidly from each other by a partition, and the heat conductors extend from the exhaust air duct through the partition into the supply air duct, wherein the exhaust air duct is open towards the condensate collector, and wherein the heat collectors are located in the exhaust air duct with an interstice between adjacent heat collectors, and the heat conductors are connected thermally to the heat collectors, wherein the interstices are open towards the condensate collector.
- 2.** The stretching unit according to claim 1, wherein the exhaust air duct defines a flow path for the air flow of the exhaust air, wherein the condensate collector is located outside of the flow path.
- 3.** The stretching unit according to claim 1, wherein the condensate collector is located at least in part below the exhaust air duct.
- 4.** The stretching unit according to claim 1, wherein the condensate collector is located at least in part below the flow path.
- 5.** The stretching unit according to claim 1, wherein the exhaust air duct comprises a condensate outlet opening that is part of the condensate collector.
- 6.** The stretching unit according to claim 1, wherein the condensate collector comprises a tray.
- 7.** The stretching unit according to claim 1, wherein the heat recovery system comprises a condensate drain and the condensate collector is connected fluidly to the condensate drain.
- 8.** The stretching unit according to claim 1, wherein the heat collectors are plate-shaped and the heat conductors extend through the heat collectors.
- 9.** The stretching unit according to claim 1, wherein the heat collectors run at least one of parallel to each other or parallel to the partition.
- 10.** The stretching unit according to claim 1, wherein the heat conductors run horizontally or are arranged at an angle of between 0° and 10° to the horizontal direction.
- 11.** The stretching unit according to claim 10, wherein the section of the heat conductor in the exhaust air duct is situated lower than that section in the supply air duct.
- 12.** The stretching unit according to claim 1, wherein the interstices comprise a direction of air flow and a direction of condensate flow, wherein the direction of air flow and the direction of condensate flow run parallel to each other.
- 13.** The stretching unit according to claim 12, wherein the direction of air flow and the direction of condensate flow run vertically or are arranged at an angle of between 0° and 10° to the vertical direction.
- 14.** The stretching unit according to claim 1, wherein the interstices comprise a direction of air flow and a direction of condensate flow, wherein the direction of air flow and the direction of condensate flow run perpendicular to each other.
- 15.** The stretching unit according to claim 14, wherein at least one of the direction of air flow runs horizontally or the direction of condensate flow runs vertically or is arranged at an angle of between 0° and 10° to the vertical direction.
- 16.** The stretching unit according to claim 1, wherein the heat conductors are heat-transfer devices.
- 17.** The stretching unit according to claim 1, wherein the oven comprises at least one of a supply air inlet which is connected fluidly to the supply air duct of the heat exchanger or an exhaust air outlet that is connected fluidly to the exhaust air duct of the heat exchanger.
- 18.** The stretching unit according to claim 1, wherein the heat recovery system comprises a heating register.
- 19.** The stretching unit according to claim 18, wherein the heating register is arranged fluidly between the supply air duct of the heat exchanger and the supply air inlet of the oven.
- 20.** A film production system comprising at least one stretching unit as well as at least one of a

further stretching unit, an extrusion unit, a cast rolling system, a draw roller unit or a winder unit, wherein the stretching unit comprises an oven and a heat recovery system, wherein the heat recovery system comprises a condensate collector and a heat exchanger with an exhaust air duct for exhaust air from the oven, a supply air duct for supply air to the oven, a plurality of heat conductors and a plurality of heat collectors, wherein the supply air duct and the exhaust air duct are separated fluidly from each other by a partition, and the heat conductors extend from the exhaust air duct through the partition into the supply air duct, wherein the exhaust air duct is open towards the condensate collector, and wherein the heat collectors are located in the exhaust air duct with an interstice between adjacent heat collectors, and the heat conductors are connected thermally to the heat collectors, wherein the interstices are open towards the condensate collector.
